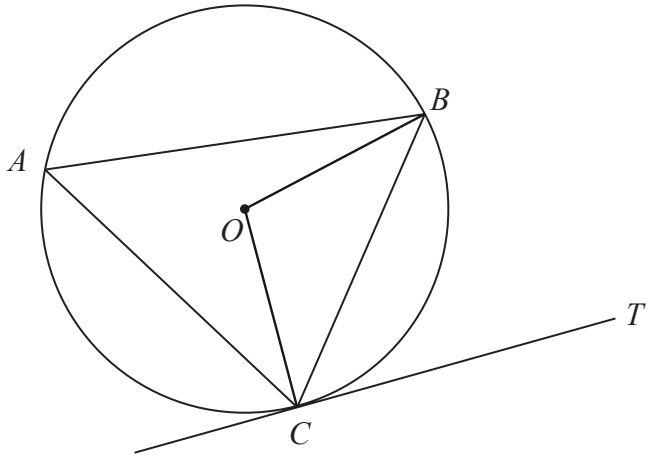
17



A, B and C are points on a circle, centre O.
CT is the tangent to the circle at C.

Prove that angle BAC = angle BCT

Let angle BAC = a. Join OB and OC.

Angle BOC = 2a: the angle at the centre is twice

the angle at the circumference on the same arc.

1 mark
Centre angle is twice a.

OB = OC (radii), so triangle OBC is isosceles.

Angle OCB = (180° – 2a) ÷ 2 = 90° – a.

1 mark
Use equal radii and base angles.

Base angles are equal; triangle angles total 180°.

Angle OCT = 90°: radius is perpendicular to tangent.

Angle BCT = 90° – (90° – a) = a = angle BAC.

1 mark
Establish the required equality.

1 mark
Full matching reasons above.

(Total for Question 17 is 4 marks)